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Bisphenol A(BPA), BPS and BPB-induced oxidative stress and apoptosis mediated by mitochondria in human neuroblastoma cell lines

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A R T I C L E I N F O

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A B S T R A C T

The analogues of biphenol A (BPA), including bisphenol S (BPS) and bisphenol B (BPB), are commonly used to replace the application of BPA in containers and wrappers of daily life. However, their safeties are questioned due to their similar chemical structure and possible physiological effects as BPA. To investigate the neurotoxic effects of BPA, BPS, and BPB as well as their underlying mechanism, IMR-32 cell line from male and SK-N-SH cell line from female were exposed respectively to BPA, BPS and BPB with concentrations of 1 nM, 10 nM, 100 nM, 1 μM, 10 μM, and 100 μM for 24 h. Additionally, 24 h exposure of BPA combining epigallocatechin gallate (EGCG) (4 μM and 8 μM for IMR-32 and SK-N-SH respectively) were conducted. Results demonstrated that BPs exposure could promote reactive oxygen species production and increase level of malondialdehyde (MDA) while decrease levels of superoxide dismutase (SOD). Intensive study revealed that after exposure to BPA mitochondrial membrane potential (MMP) dropped down and the protein expression levels of Bak-1, Bax, cytochrome c and Caspase-3 were up-regulated but Bcl-2 were down-regulated significantly. Moreover, apoptosis rate was raised and cell activity declined remarkably in the neuroblastoma cells. All the effects induced by BPA could be alleviated by the adding of EGCG, which similar alleviations could be inferred in IMR-32 and SK-N-SH cells induced by BPS and BPB. Furthermore, BPS showed lower neurotoxic effects compared to BPA and BPB. Interestingly, the neurotoxic effects of BPA on IMR-32 cells were significantly higher than those on SK-N-SH cells. In conclusion, the results suggested that BPA, BPS and BPB could induce oxidative stress and apoptosis via mitochondrial pathway in the neuroblastoma cells and male is more susceptible to BPs than female.

1. Introduction

Bisphenol A (BPA), a lipophilic and endocrine disrupting chemicals (EDCs) with high production volume, is used as an additive in manufacturing polycarbonate plastics and epoxy resins (Kohda et al., 2012; Wang et al., 2019a, 2019b; Zhang et al., 2011). Many daily utensils contain BPA commonly, including food and water containers and packages, tooth solid sealing agents, baby bottles, infusion bags and toys (Hoekstra et al., 2013; Makris et al., 2013). Global BPA production had exceeded 4.6 million tons in 2012 and is expected to grow at an annual rate of 4.6% from 2013 to 2019 (Wang et al., 2017). Large quantities of BPA are released into environment during its production and using. Consequently, human is inevitably and passively exposed to BPA. As a result, BPA is detected in human amniotic fluid, blood, breast milk, placenta, sweat and urine (Shi et al., 2017a, 2017b; Yu et al., 2019).

Previous researches suggested that prenatal and perinatal BPA exposure induced precancerous and cancerous lesions, and altered mammary gland development in rats (Dhimolea et al., 2014; Durando et al., 2007). Additionally, BPA is reported with hepatotoxicity and neurotoxicity (Hassan et al., 2012; Senyildiz et al., 2017). For example, BPA exposure led to substantial DNA damage in HepG2 cells (Fic et al., 2013) and impaired cognitive, learning and memory abilities (Braun et al., 2017; Zhang et al., 2019). Epidemiological studies imply the associations of BPA exposure with the increasing incidence of cardiovascular disease, childhood obesity, and type II diabetes. Considering the

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adverse health risk, many countries and areas issued regulations to ban or restrict the additive of BPA in the production of many containers, including baby bottles. To evade the restriction, BPA analogues, such as bisphenol S (BPS) and bisphenol B (BPB) with similar chemical structures as BPA, are gradually replacing the BPA in baby products which are labeled with BPA-free. The global production and application of bisphenol analogues (BPs) are rising (Konno et al., 2004). Monitoring survey on common population indicated that daily dietary intakes of BPB and BPS for adults in the United States were 0.35 and 1.31 ng/kg, respectively (Liao et al., 2013). However, the safety of BPA analogues is not well known. Some evidences suggest that the toxic effects of BPs are equal or even greater than BPA (Moreman et al., 2017; Tucker et al., 2018). Kinch et al. found that BPS alters brain development and behavior (Kinch et al., 2015), and BPB was found to have endocrine disrupting effects with more leaching into canned foods compared to BPA (Ikhlas et al., 2019; Tucker et al., 2018).

Previous studies revealed that the promoted apoptosis was one of the major toxicity mechanisms of EDCs (Chen et al., 2019). The action of ROS generated by environmental pollutants in apoptosis can be validated by ROS scavengers. For instance, N-acetylcysteine, a free radical scavenger, could inhibit the pro-apoptotic effect of BPA (Peerapanyasut et al., 2019), polystyrene microplastics (Xie et al., 2019) and aniline (Wang et al., 2016). It is well-known that mitochondrion is not only the source of ROS generation, but also the target of oxygen free radical (Orrenius et al., 2007). Excessive ROS can reduce the mitochondrial membrane potential (MMP), change the mitochondrial membrane permeability and finally produce apoptosis by releasing apoptotic factors (Yang et al., 2019), which includes cysteinyl aspartate specific proteinase 3 (Caspase-3), Bcl-2-Associated X (Bax), Recombinant Bcl-2 Antagonist/Killer 1 (Bak1), Recombinant B-Cell Leukemia/Lymphoma 2 (Bcl-2) and cytochrome C (Cyt C) etc. *In vitro*, BPA can directly affect human sperm integrity by inducing pro-oxidative/apoptotic mitochondrial dysfunction (Barbonetti et al., 2016). Our previous studies found that BPA, BPS and BPB induced generation of ROS and cell apoptosis in hippocampal cell line (Pang et al., 2019). However, the intensive mechanism is not clear yet.

In addition, gender dependent outcomes of BPA exposure are found in numerous investigation. In order to survey the sex-specific neurotoxic effect of BPA and its analogues, two neuroblastoma cell lines of different genders were selected as the research objects. The SK-N-SH cell line was derived from a girl metastatic neuro blastoma (Biedler et al., 1973), and the IMR-32 cell line was from tumor in the lower abdomena of a 13-month-old baby boy (Tumilowicz et al., 1970), which are commonly used as cellular model for research (Chen et al., 2020; Tang et al., 2019a, 2019b). Epigallocatechin gallate (EGCG) is a phenolic antioxidant compound extracted from green tea that can alleviate/reduce oxidative stress (Han et al., 2014; Singh et al., 2016). In the present study, neuroblastoma cells were cultured in medium in the absence and presence of BPA, BPS and BPB, in EGCG and the combination of BPA and EGCG to investigate BPA-induced apoptosis via mitochondria-mediated caspase-dependent pathway.

2. Materials and methods

2.1. Cell culture and treatment

Two human neuroblastoma cell lines (i.e, IMR-32 and SK-N-SH from male and female, respectively) were obtained from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China). They were cultured in minimal essential medium (MEM) (Gibco, 41500034) added with 10% fetal bovine serum (FBS) (Biological Industries, 04-001-1ACS), 1% penicillin-streptomycin (Gibco, 15140163) and 1% sodium pyruvate (Solarbio, P8380) at 37 °C in an incubator with 5% CO_2. Then they were cultured in medium in the absence and presence of bisphenols (BPs), in EGCG, as well as in the combination of BPA and EGCG, respectively. BPA, BPS and BPB (all with 97–99% purities) were dissolved in dimethyl sulfoxide (DMSO Sigma, D4540). Concentrations of DMSO in all samples were less than 0.1%. IMR-32 and SK-N-SH cells were seeded on well dishes and attached overnight. After overnight incubation, cells were treated with different concentrations of BPA or its analogues (i.e., 1 nM, 10 nM, 100 nM, 1 μM, 10 μM and 100 μM) in MEM medium (Sigma, M3024) containing 10% carbon adsorbed FBS and without phenol red for 3, 6, 12, 24, 48 and 72 h, respectively. In addition, IMR-32 and SK-N-SH cells were treated with EGCG for 24 h and the treated cells were divided into 4 groups, including the control group only containing 0.1% DMSO, the BPA groups with the adding of 10 μM BPA or 100 μM BPA, the EGCG groups with the adding of 4 μM EGCG or 8 μM EGCG, and the BPA + EGCG groups with the adding of 10 μM BPA and 4 μM EGCG, or 100 μM BPA and 8 μM EGCG.

2.2. Determination of reactive oxygen species (ROS), superoxide dismutase (SOD) and malondialdehyde (MDA) levels

Measurement of ROS level was conducted by 2,7-dichlorofluorescein diacetate (DCFH-DA) assays (Beyotime, China). Cells were incubated with 4 μM of DCFH-DA for 20 min at 37 °C under dark. Then, cells were washed with phosphate buffer saline (PBS) and analyzed immediately by flow cytometry with the excitation and emission wavelengths of 488 and 525 nm, respectively.

Intracellular SOD activity and MDA content were assayed by commercially available kits (Nanjing Jiancheng Bioengineering Institute, China).

2.3. Determination of mitochondria membrane potential (MMP)

MMP was determined by MMP assay kit with JC-1 (Beyotime, China). 6×10^5 cells of IMR-32 and SK-N-SH were freshly collected into a microcentrifuge tube on the day of assay. Cells were resuspended in the mixture of 0.5 mL MEM and 0.5 mL JC-1 and incubated in a tissue culture incubator (at 37 °C and with 5% CO_2) for 20 min. Then MMP was assessed by flow cytometer (FACS Aria III, BD Biosciences, USA) at the 590 nm excitation wavelength and 520 nm emission wavelength.

2.4. Apoptosis analysis

Cell apoptotic rate was measured by Annexin V-FITC Apoptosis Detection kit (Beyotime Biotechnology, Shanghai, China). IMR-32 and SK-N-SH cells were planted in 24-well plates and incubated overnight. After 24 h treatments, cells were harvested by trypsinization, resuspended in 400 μL of buffer and followed by a centrifugation at 1000×*g* for 5 min. Consequently, cells were resuspended in 195 μL of Annexin V-fluorescein isothiocyanate (FITC) binding buffer. Then, cells were double-stained with 5 μL of Annexin V-FITC and 10 μL of propidium iodide (PI) for 15 min at room temperature in dark. All specimens were subjected into a flow cytometer at the 488 nm excitation wavelength, the 520 nm emission wavelength of FITC and the 620 nm emission wavelength of PI to discriminate cell apoptosis.

2.5. Western blotting analysis

IMR-32 and SK-N-SH cells were harvested from the control and BPA-treated groups and homogenized in cold RIPA buffer (Beyotime, China) containing protease inhibitors (Beyotime, China). Lysates were subsequently collected and centrifuged at 12,000 rpm for 10 min at 4 °C and the total protein concentrations were measured using a BCA protein assay kit (KeyGEN BioTECH, China). Total protein samples were diluted in the 5 × SDS-PAGE loading buffer and boiled for 5 min. Then, 40 μg of proteins were resolved in the 10% SDS-PAGE and transferred onto 0.22 or 0.45 μm polyvinylidene difluoride (PVDF) membranes (Millipore, USA). After a subsequent sealing with 5% skimmed milk powder diluted with Tris-Borate Tween-20 buffer (TBST) for 2 h at room temperature, the membrane was incubated for 16 h with the diluted first antibodies,

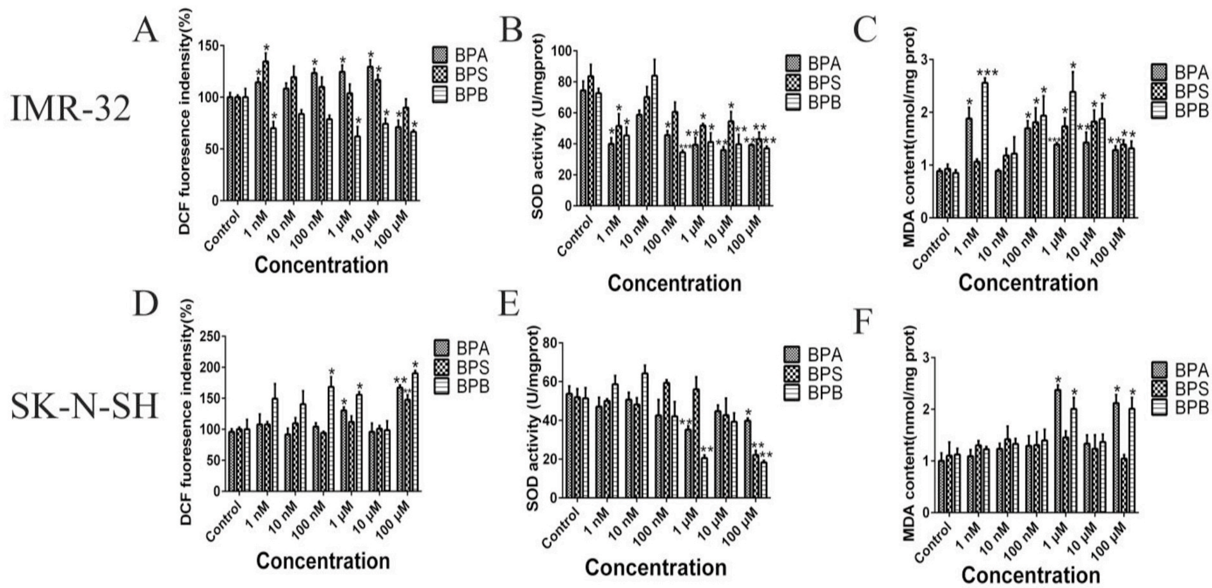

Fig. 1. ROS, SOD and MDA levels in IMR-32 and SK-N-SH cells after different concentrations exposure to BPA, BPS and BPB for 24 h. Effects on ROS levels of BPA, BPS and BPB treatments(A, D). Effects on SOD activities of BPA, BPS and BPB treatments (B, E). Effects on MDA contents of BPA, BPS and BPB treatments (C, F). The results were achieved based on at least three repeated experiments. * indicates the significant difference between the control and treated groups (*$p < 0.05$, **$p < 0.01$ and ***$p < 0.001$).

including β-actin (1:1000, Cell signaling Technology, USA, # 8457S), Bak1 (1:1000, Cell signaling Technology, USA, # 22105S), Bax (1:1000, Cell signaling Technology, USA, # 2772S), Caspase-3 (1:1000, Cell signaling Technology, USA, # 9662S), Bcl-2(1:1000, Sigma, USA, # SAB4500003) and Cyt c (1:1000, Sigma, USA, # SAB4502234). Then after washing with TBST and incubating with horseradish peroxidase (HRP)-conjugated secondary antibodies for 1 h at room temperature, the protein levels were measured with a ChemiDoc Touch Imager (Bio-Rad, USA) after adding electrochemiluminescence liquid. All proteins levels were normalized to β-actin protein level and analyzed by Image Lab software.

2.6. Cell viability assay

IMR-32 and SK-N-SH cells were incubated in 96-well plates at a density of 2×10^4 cells per well. Then BPs were added when cells had grown to cover 70% of plates. Cell viability was assessed by the CCK-8 assay (APExBIO, USA) according to the manufacturer's protocols. After 12, 24, 48 and 72 h of BPs treatments, 10 μL of CCK-8 was added to each well. Then, IMR-32 and SK-N-SH cells were incubated for 4 h at 37 °C. At last, a microplate reader (PerkinElmer; Enspire, USA) was used to get their absorbances. Cell viability was calculated according to the following formula: cell viability (%) = [(As−Ab)/(Ac−Ab)] × 100%, where As, Ab and Ac represent the absorbances of the treated, untreated and blank groups at 450 nm, respectively.

2.7. Statistical analysis

All experiments were independently repeated at least three times and data was expressed as the mean ± standard deviation (SD). All statistical analysis used GraphPad Prism 6.0 (GraphPad Inc., La Jolla, CA, USA) and SPSS version 18.0 (SPSS Inc., Chicago, IL, USA). Variance (ANOVA) was used to test the difference between groups and significance was set at the level of $p < 0.05$.

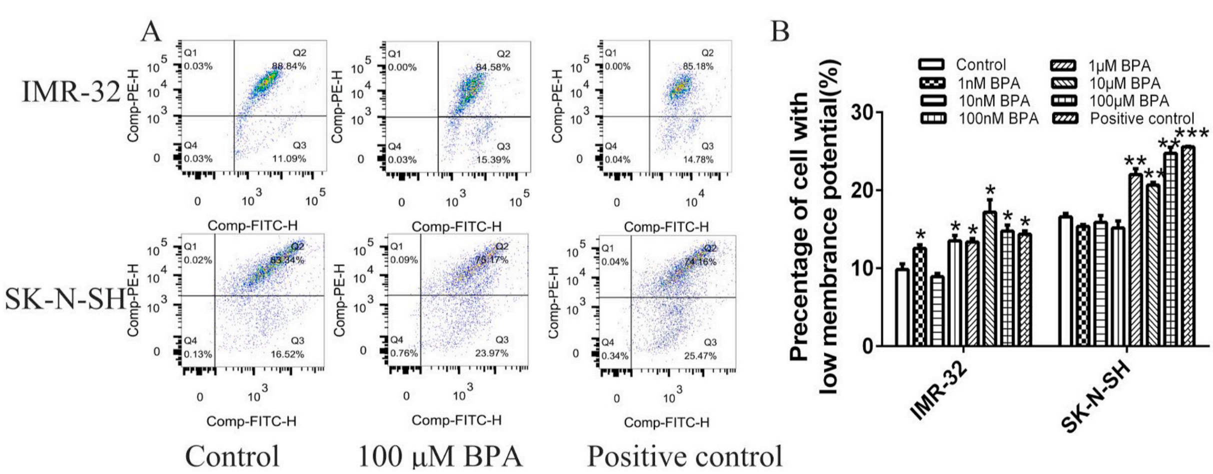

Fig. 2. Effects on MMP in IMR-32 and SK-N-SH cells after BPA exposure for 24 h treatment. (A) Flow cytometry figures of MMP. (B) Percentage of cell with low MMP. The results were achieved based on at least three repeated experiments. "*" indicates the significant difference between the control and treated groups (*$p < 0.05$, **$p < 0.01$ and ***$p < 0.001$).

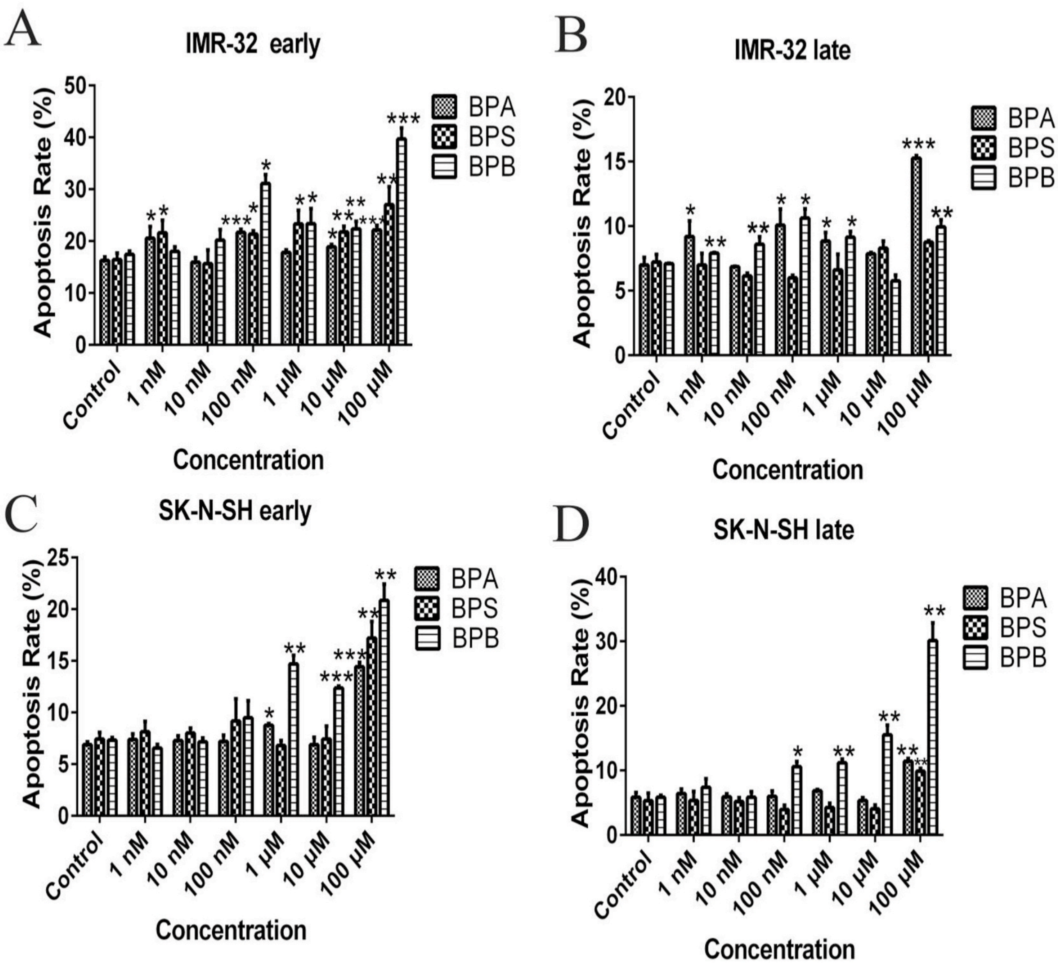

Fig. 3. Apoptosis of IMR-32 and SK-N-SH cells exposed to different concentrations of BPA, BPS and BPB for 24 h. Early apoptosis rates after BPs exposure (A, C). Late apoptosis rates after BPs exposure (B, D). The results were achieved based on at least three repeated experiments. * indicates the significant difference between the control and treated groups (*$p < 0.05$, **$p < 0.01$ and ***$p < 0.001$).

3. Results

3.1. Cell oxidative stress induced by BPs for IMR-32 and SK-N-SH cells

Oxidative stress is considered the initialized step during the development of oxidative damage and apoptosis (Tang et al., 2019a, 2019b). Thus, ROS and SOD, MDA levels were measured. After 24 h exposure, ROS was emitted again in the BPs treated groups of IMR-32 cell (i.e., 1 nM, 100 nM, 1 μM, 10 μM and 100 μM of BPA, 1 nM and 10 μM of BPS and 1 nM, 1 μM, 10 μM and 100 μM of BPB) (Fig. 1A) ($p < 0.05$ or $p < 0.01$). Similarly, 1 μM and 100 μM of BPA, 100 μM of BPS, 100 nM, 1 μM and 100 μM of BPB treatments also led to the significant release of ROS again in SK-N-SH cell (Fig. 1D) ($p < 0.05$ or $p < 0.01$), which suggesting a different trend between genders. ROS were measured after 3, 6, 12, 24 and 48 h treatments (Supplementary Fig. 1), the production of ROS in both cells showed a significant non-monotonic behavior after 24 h treatment. Hence, the determinations of SOD, MDA, MMP and apoptosis test were conducted after 24 h treatment.

As shown in Fig. 1B–C, at the 24th hour treatment, the activities of SOD in IMR-32 cells were obviously decreased while the contents of MDA were significantly elevated in all BPs treated groups with the exception of 10 nM BPA treated group, 1 nM, 10 nM and 100 nM BPS treated groups and 10 nM BPB treated group ($p < 0.05$ or $p < 0.01$). Moreover, for SK-N-SH cells, only higher exposure treatments (i.e., 1 μM and 100 μM BPA and BPB and 100 μM BPS) could significantly reduce SOD; In the meanwhile, 1 μM and 100 μM of BPA and BPB exposure increased MDA levels (Fig. 1E-F) ($p < 0.05$ or $p < 0.01$).

3.2. Effects of BPA on MMP in IMR-32 and SK-N-SH cells

After 24 h incubation, results showed that the numbers of IMR-32 cells in the Q3 fraction were substantially enhanced from 9.83% to 12.53%, 13.52%, 13.36%, 17.19%, 14.76% and 14.35% after the treatments with low and high BPA (i.e.,1 nM, 100 nM, 1 μM, 10 μM, 100 μM and positive control), respectively ($p < 0.05$ or $p < 0.01$) (Fig. 2A–B). For SK-N-SH cells, 1 μM–100 μM BPA treatments induced the significantly MMP loss compared with the control one. However, 1 nM, 10 nM and 100 nM BPA exposure did not lead to significant effects on the MMP of SK-N-SH cells.

3.3. Apoptosis induced by BPs exposure in IMR-32 and SK-N-SH cells

Apoptosis in IMR-32 and SK-N-SH cells after BPA, BPS and BPB treatments were examined using the PI/annexin V-FITC double staining and fluorescence intensity by flow cytometry. As shown in Fig. 3A–B, BPA (i.e., 1 nM, 100 nM, 10 μM and 100 μM), BPS (i.e., 1 nM and from 100 nM to 100 μM) and BPB (i.e., form 100 nM to 100 μM) treatments for 24 h significantly elevated the portions of early apoptotic cells for IMR-32 cells ($p < 0.05$ or $p < 0.01$). Moreover, except for BPS, BPA and BPB could also increase the ratios of late apoptotic cells for IMR-32 cells ($p < 0.05$ or $p < 0.01$). Similarly, in addition to 1 μM and 100 μM BPA, 100 nM-100 μM BPB and 100 μM BPS treatments led to the apoptosis of SK-

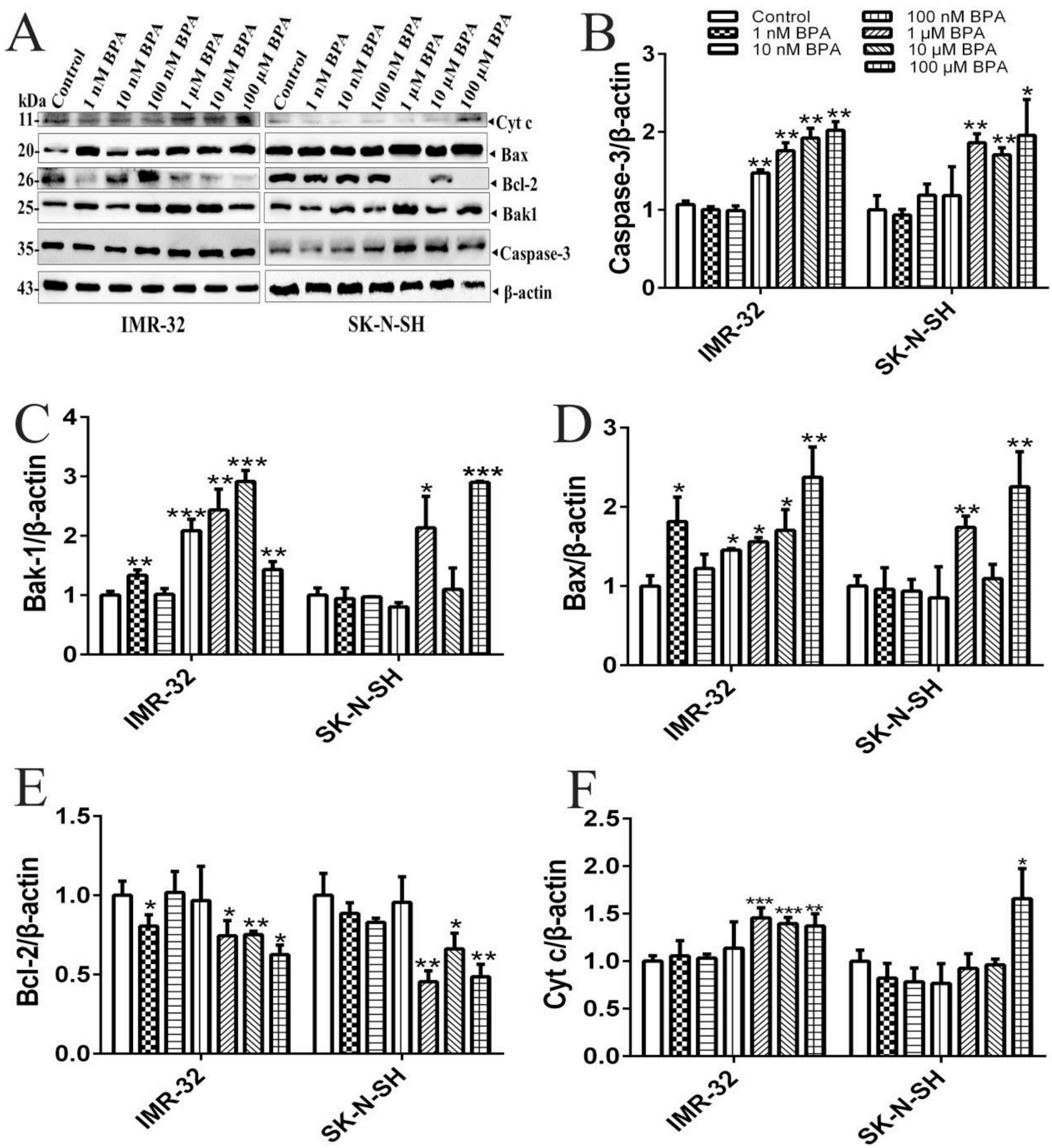

Fig. 4. Effects on Caspase-3, Bak1, Bax, Bcl-2 and Cyt c protein levels in IMR-32 and SK-N-SH cells after different concentrations exposure to BPA for 24 h. Protein levels of Caspase-3, Bak1, Bax, Bcl-2 and Cyt c after different concentrations treatments of BPA (A). Protein levels of Caspase-3, Bak1, Bax, Bcl-2 and Cyt c compared with β-action (B–F). The results were achieved based on at least three repeated experiments. * indicates the significant difference between the control and treated groups (*$p < 0.05$, **$p < 0.01$ and ***$p < 0.001$).

N-SH cells, either ($p < 0.05$ or $p < 0.01$) (Fig. 3C–D).

3.4. *Expression of apoptosis proteins in IMR-32 and SK-N-SH cells induced by BPA*

To confirm our hypothesis that BPA induces apoptosis through mitochondria mediated pathway, the levels of mitochondria mediated pathway-associated proteins, such as Caspase-3, Bak1, Bcl-2, Bax and Cyt c were determined. Results showed that, for IMR-32 cells, levels of Caspase-3, Bak1, Bax and Cyt c in the BPA treated groups were significantly increased; However, Bcl-2 level was significantly reduced compared to the control group with the exception of 10 nM BPA treated group ($p < 0.05$ or $p < 0.01$) (Fig. 4A–F). Similarly, for SK-N-SH cells, expressions of Caspase-3, Bak1, Bax and Cyt c in the groups treated with higher concentration of BPA (i.e., 1–100 μM BPA or 1 μM and 100 μM BPA) were significantly increased and Bcl-2 level in the 1–100 μM BPA treated groups were significantly decreased compared to the control one ($p < 0.05$); whereas no significant differences were observed between the other treated groups and the control one ($p > 0.05$).

3.5. *5 Morphologic alterations and viabilities of IMR-32 and SK-N-SH cells*

After 24 h exposure to different concentrations of BPA,

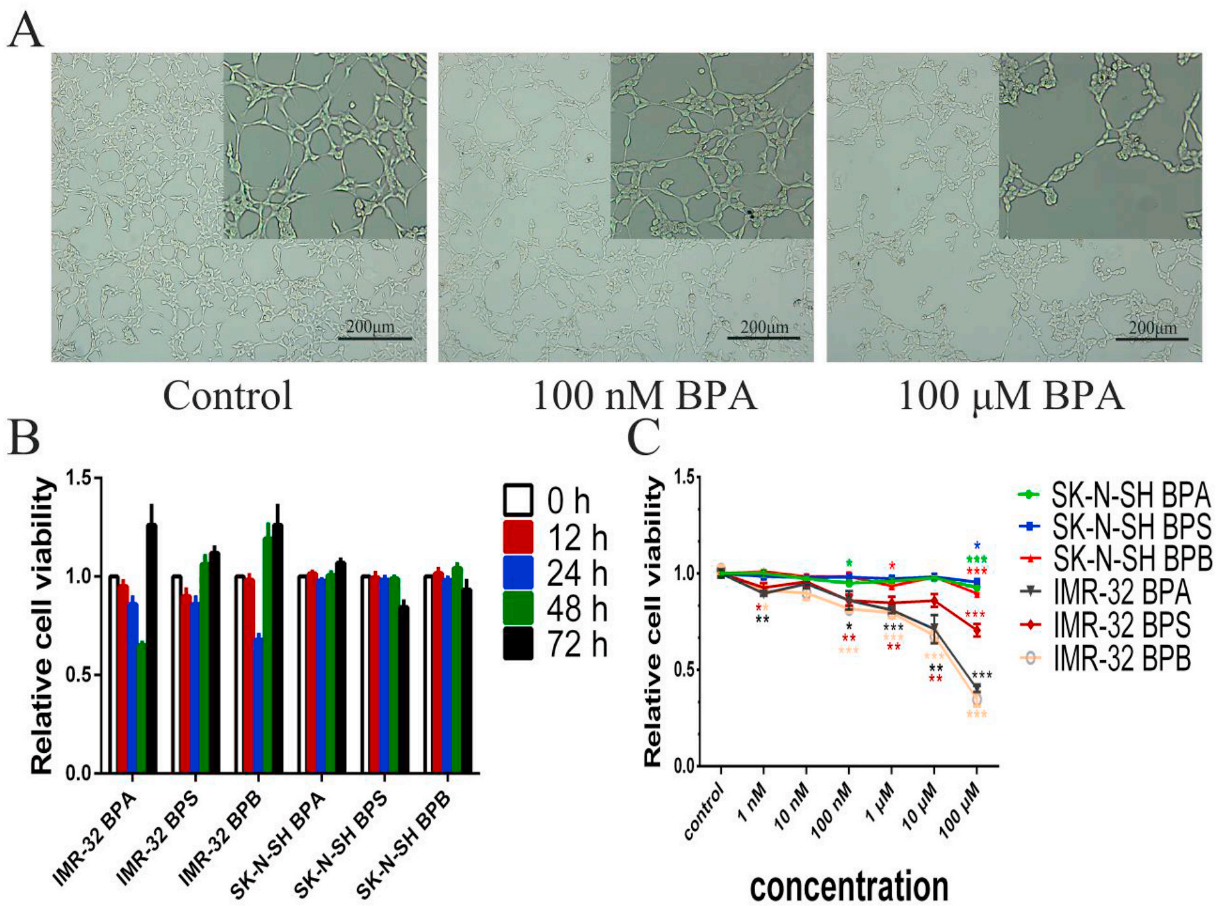

Fig. 5. Morphologic alterations and cell viabilities of IMR-32 and SK-N-SH cells. Morphological changes of IMR-32 cell under a normal microscope (100x magnification) (A). Effects of 10 μM BPA, BPS and BPB on the viabilities of IMR-32 and SK-N-SH cells for 12 h, 24 h, 48 h and 72 h treatments respectively (B). Effects of different concentrations of BPA, BPS and BPB on the viabilities of IMR-32 and SK-N-SH cells for 24 h treatments (C). The results were achieved based on at least three repeated experiments. * indicates the significant difference between the control and treated groups (*$p < 0.05$, **$p < 0.01$ and ***$p < 0.001$).

morphological alterations of cells were observed under a light microscope (Leica, DM6 Germany). As shown in Fig. 5A, BPA induced soma rounded up and the intercellular connections disappeared due to the loss of dendrite in IMR-32 cells. In the meantime, SK-N-SH cells showed the same morphological alterations.

At a 24 h exposure to BPs, cell viabilities of IMR-32 were decreased with the treatments of both low and high BPs (i.e., 1 nM, 100 nM, 1 μM, 10 μM and 100 μM BPA and BPB, in addition to 1 nM, 100 nM, 1 μM, 10 μM and 100 μM BPS) ($p < 0.05$); While cell viabilities of SK-N-SH were significantly decreased by higher level treatments of BPs (i.e., 100 nM and 100 μM BPA, 100 μM BPS, 1 μM and 100 μM BPB) (Fig. 5C). Results of Fig. 5B indicated that for IMR-32 cells, cell viabilities were significantly decreased after 12, 24 and 48 h BPA treatments, whereas were increased after 72 h BPA treatment. Similar trends were also observed for BPS and BPB treatments, suggesting the non-monotonic dose-response relationships between the BPs concentrations and the viabilities of neuroblastoma cells.

3.6. EGCG reduces BPA-induced cytotoxicity in IMR-32 and SK-N-SH cells

We examined whether ROS production plays an important role in BPA-induced apoptosis by adding EGCG (an antioxidant). Since BPA increased the formation of ROS, a common antioxidant (i.e., EGCG) was added to the BPA treated groups to study its slowing-down effect on BPA-induced cytotoxicity. 24 h treatment was selected due to its significant effect. ROS was measured after neuroblastoma cells had been co-incubated with EGCG and BPA for 24 h. 10 μM and 100 μM BPA were used due to their distinctive cytotoxic effects achieved in ROS and apoptosis experiments. 4 μM and 8 μM EGCG were selected as the experimental doses because they were the maximum nontoxic doses to IMR-32 and SK-N-SH cells, respectively. Results showed that EGCG reduced BPA-induced ROS generation in IMR-32 and SK-N-SH cells at the 24th hour treatment ($p < 0.05$) (Fig. 6A). Furthermore, EGCG could significantly enhance the activity of SOD and reduce the content of MDA in BPA-treated cells (i.e., 10 μM BPA treated group for IMR-32 and 100 μM BPA treated for SK-N-SH) ($p < 0.05$). No significant difference was observed between the control and the EGCG groups ($p > 0.05$) (Fig. 6C–D).

10 μM and 100 μM of BPA were chosen to explore their effects on the apoptosis of IMR-32 and SK-N-SH cells. Moreover, EGCG and BPA co-treatment for 24 h decreased the portions of early apoptotic cells from 31.2% to 24.65% in IMR-32 cells and from 14.43% to 11.4% in SK-N-SH cells; the ratios of late apoptotic cells were also decreased from 9.07% to 6.33% and from 13.7% to 11.81% after the treatments with 10 μM and 100 μM of BPA, respectively ($p < 0.01$) (Fig. 6E–F). EGCG significantly decreased the expression of apoptosis-related proteins (such as Caspase-3, Bak1, Bax and Cyt c) and up-regulated the expression of anti-apoptosis protein (i.e., Bcl-2) (Fig. 6G–H).

4. Discussion

Our previous research revealed that BPs exerted neurotoxicity by impairing cell viability, producing ROS and accelerating cell apoptosis

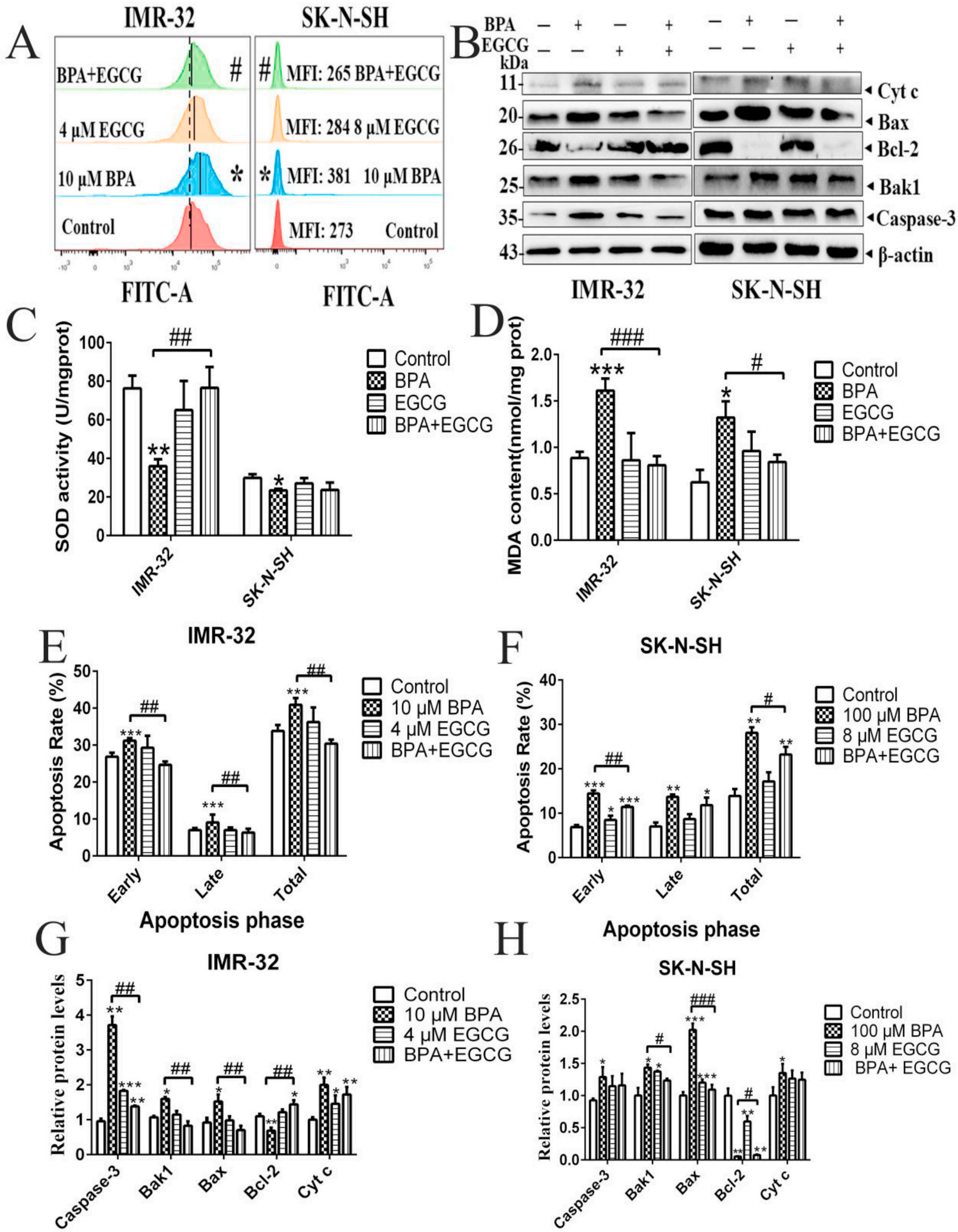

Fig. 6. Effects of BPA (i.e., 10 μM and 100 μM) and EGCG (i.e., 4 μM and 8 μM) on oxidative stress and apoptosis in the IMR-32 and SK-N-SH cells for 24 h treatments. ROS production. (B) Protein levels of Caspase-3, Bak1, Bax, Bcl-2 and Cyt c (A). SOD activity and MDA content(C, D). Apoptosis rate of IMR-32 and SK-N-SH cells (E, F). Relative protein levels of Caspase-3, Bak1, Bax, Bcl-2 and Cyt c (G, H). The results were achieved based on at least three repeated experiments. * indicates the significant difference between the control and treated groups (*$p < 0.05$, **$p < 0.01$ and ***$p < 0.001$). "#" indicates the significant difference between two corresponding groups (#$p < 0.05$, ##$p < 0.01$ and ###$p < 0.001$).

in HT-22 cell line (Pang et al., 2019), but the underlying mechanism is unclear. In this study, we performed further exploration for the mechanism of neurotoxicity with two kinds of human neuroblastoma cells, mainly focusing on the association among oxidative stress, mitochondria and apoptosis.

4.1. Oxidative stress induced by BPs exposure

ROS is the reactive metabolites of oxygen that could be radicals (like superoxide anion and hydroxyl anion) or no-radicals, which have a pivotal role in physiological and pathological processes. Excessive ROS results in oxidative stress, which causes damages of lipids and proteins and then undergoes biomembranes disrupting and enzymes inactivating, finally leads to pathological changes of cell structural and functional disorder (Wei et al., 2019; Schieber et al., 2014). Imbalance of ROS production and antioxidant defenses is the characterized for oxidative stress. The enzyme system composed of specific antioxidation-related enzymes is the main antioxidant system (Bresciani et al., 2015). SOD is regarded as the most vital antioxidant enzyme against ROS. The content of MDA, a lipid peroxidation product, reflects the severity of oxidative damage for cell.

In this study, ROS and MDA levels in both IMR-32 and SK-N-SH cells were increased after the treatments of BPA, BPS and BPB, respectively. But SOD activities were decreased, which suggested that these three kinds of BPs chemicals could break the balance of oxidation-antioxidant system in neuroblastoma cells to trigger oxidative stress. The result was consistent with the previous reports (Maćczak et al., 2017; Pandey et al., 2011). Interestingly, non-monotonic dose-effect relationship was observed between the concentrations of BPs and effects of oxidative stress, which might be attributed to their characteristics of environmental endocrine disruptors (Vandenberg, 2014).

4.2. The decrease of MMP induced by BPA increased the apoptosis and decreased the viability of neuroblastoma cells

It is well known that mitochondria play an important role in the formation of free radicals in living cells (Circu et al., 2010) and is a main source generating ROS (Cid-Castro et al., 2018). Many studies suggested that the apoptotic signal pathway is mediated by mitochondria with its critical role in cell survival for generation of ATP via oxidative phosphorylation (Wang et al., 2020a, 2020b; Yang et al., 2019). In addition to energy supplies, mitochondria regulate cell death pathways involving the integration and circulation of death signals initiating, such as oxidative stress and apoptosis (Green, 2004; Sinha et al., 2013).

MMP is an important biomarker for apoptosis induced by oxidative stress, and the decrease of MMP is considered to be a later event in the apoptosis pathway (Ding et al., 2012). In our research, the decrease of MMP in the BPA-treated group accompanied with the increase of apoptotic rate was observed in neuroblastoma cells. Apoptosis are considered to be regulated by various pro-apoptotic proteins and anti-apoptotic proteins. Bcl-2 are a protein family that lurk on the surface of mitochondria and introduce mitochondria into the cell death pathway (Cory et al., 2003). Bax and Bak, are two apoptotic proteins of Bcl-2 family, which are transformed from harmless monomers into deadly oligomers and formed pores in the mitochondrial outer membrane. The pores are conduits for pro-apoptotic factors to translocate to the cytoplasm (Westphal et al., 2014), and Cyt c is one of those pro-apoptotic factors that would instigate proteolytic cascade to dismantle the cell (Liu et al., 1996). In a word, the anti- and pro-apoptotic protein markers (Bcl-2 and Bax) play important roles in the cell death pathway for mitochondrial apoptosis. In that pathway, once cells are stimulated, Bcl-2 family proteins in cytoplasm will transfer to the mitochondrial membrane to active downstream proteins including Caspase-3, a principal effector of apoptotic cascades that would result in neuro-degeneration (Louneva et al., 2008). Our results suggested that BPA could induce apoptosis in neuroblastoma cells through the caspase-dependent pathway related to oxidative stress mediated by mitochondria.

As a phenolic antioxidant (Zhong et al., 2012), EGCG was added in neuroblastoma cells to scavenge ROS. Comparing the BPA and the BPA + EGCG groups, we could observe that EGCG suppressed the oxidative stress induced by BPA via increasing SOD activity and decreasing ROS and MDA levels. Additionally, EGCG reduced the elevation of apoptotic rate induced by BPA, accompanying with reducing the expressions of Bak-1, Bax, Cyt c and Caspase-3 proteins and increasing the expression of Bcl-2 protein caused by BPA. The results of EGCG experiment made a further verification that oxidative stress involves in the BPA-induced apoptosis via mitochondrial pathway.

Cell proliferation and apoptosis always maintain a dynamic balance (Shi et al., 2017a, 2017b). Lee et al. observed that high-dose BPA suppressed proliferation of HT-22 cells, but low-dose BPA promoted proliferation (Lee et al., 2008). However, our study found that both low and high BPA exposure (1 nM and 100 μM) significantly decreased cell viability and inhibited the cell proliferation, accompanied by a shrinkage of cell body in neuroblastoma cells, which is consistent with our previous result (Pang et al., 2019). These phenomena may be attributed to the different exposure times and cell types.

4.3. Gender dependent effects and toxicity assessment of BPA, BPS and BPB

Neuroblastoma cell lines from different gender respond distinctively to BPA, BPS and BPB treatments. In this study, we observed the distinctive difference of protein levels induced by BPA between different neuroblastoma cells from female and male. For IMR-32 cells, both low and high BPA treatments, even 1 nM BPA treatment, upregulated the Caspase-3, Bak-1, Bax, and Cyt c protein levels and inhibited the Bcl-2 protein level. However, no significant changes of protein expression were observed in SK-N-SH cells after 1–100 nM BPA treatments, which suggesting that lower BPA levels could not influence protein levels of female neuroblastoma cell. Moreover, the cell viabilities of IMR-32 were significantly lower than those of SK-N-SH at the same treatment levels, which suggested that male neuroblastoma cells were more susceptible to BPs than female ones. In total, the toxic effects of BPA and its analogues on IMR-32 cells were significantly higher than those on SK-N-SH cells. Hence, males are more sensitive to BPs than female.

The sex-specific effects of BPA and its analogues probably own to their estrogen-like action. As an environmental estrogen, BPA can mimic

Table 1
The lowest concentrations of BPA, BPS and BPB induced significant toxic variations in IMR-32 and SK-N-SH cells ($p < 0.05$).

		ROS level (24 h)	*SOD activity (24 h)*	MDA content (24 h)	*Cell viability (24 h)*	Early apoptotic (24 h)	Late apoptotic (24 h)
IMR-32 (Male)	BPA	1 nM	1 nM	1 nM	1 nM	1 nM	1 nM
	BPS	1 nM	1 nM	100 nM	1 nM	1 nM	>100 μM
	BPB	1 nM	1 nM	1 nM	1 nM	100 nM	1 nM
SK-N-SH (Female)	BPA	1 μM	1 μM	1 μM	100 nM	1 μM	100 μM
	BPS	1 μM	100 μM	>100 μM	100 μM	100 μM	100 μM
	BPB	100 nM	1 μM	1 μM	1 μM	1 μM	100 nM

or interfere with endogenous estradiol (E2) by binding and modulating estrogen receptors (ER) (Matthews et al., 2001). In addition, BPA has been proven to strongly inhibit androgen receptor (AR) (Teng et al., 2013). Sadowski et al. revealed that male rats exposed to 400 μg/kg BPA were found to have the increased numbers of neurons and glia in the prefrontal cortex, while female rats showed no significant changes (Sadowski et al., 2014). Similar situation were also found in goldfish that BPA exposure reduces germ cell maturation in male fishes, but does not affect estrogen levels and the sex-specific gene expression in female fish (Wang et al., 2019a, 2019b). Perinatal exposure to a low dose of BPA increased anxiety-like behavior and dopamine levels and monoamine oxidase-B activity levels in the limbic and medulla oblongata of male mice, but those changes were not observed in female mice (Matsuda et al., 2012). BPA impacts on human beings include reproductive effects (Hart, 2020), mammary gland developmental problems (Osborne et al., 2015) and low sperm production (Kabir et al., 2015) etc. Given that males may be more affected by sex-hormone imbalances than females (Varshney et al., 2017), even in the same exposure conditions of BPA, men and women could manifest different health effects (Caporossi et al., 2015).

The profile lowest concentrations of BPA, BPS and BPB that induced significant toxic effects in IMR-32 and SK-N-SH cells were summarized in Table 1. Results showed that 1 nM BPA and BPB could result in ROS, inhibit cell viability and induce cell apoptosis, and only higher than 100 nM BPS can induce late apoptotic of male neuroblastoma cell (i.e., IMR-32). For the female neuroblastoma cell (i.e., SK-N-SH), only BPA levels up to microgram can lead to significant effects, such as ROS, cell viability and cell apoptosis, which are 100 or 1000 times of the BPA-treated levels in IMR-32. Compared with BPA and BPB, BPS showed lower cytotoxicity, particularly for IMR-32 cell (Fig. 5C). The differential bioactivities of BPs are thought to be linked with estrogen/androgen receptor affinity, phospholipophilicity, conformations, and receptor-ligand-binding domain specificity with different energetics (Cao et al., 2017; Li et al., 2018; Russo et al., 2018; Welshons et al., 2003). Our findings suggest that the substitutes of BPA may not be safe, which are consistent with the previous studies (Qiu et al., 2019; Wang et al., 2020a, 2020b; Yin et al., 2019).

5. Conclusions

BPA, BPS and BPB induce oxidative stress by generating excessive ROS and MDA, while apoptosis was mediated by the mitochondrial pathway involving ROS in neuroblastoma cell. Male is more susceptible to BPs than female for neuroblastoma cell. The results imply that BPS might be an optional surrogate with less neurotoxic to replace BPA.

Credit author statement

This statement is to certify that Dr. Congcong Wang completed most of experimental section, including the morphologic alteration of cells, the assay of mitochondrial membrane potential and the expression of protein. Moreover, she drafted the manuscript; Miss Jiaying He completed the ROS section; MissTongfei Xu and Hongyu Han determined the SOD and MDA levels; Miss Zhimin Zhu and Lingxue Meng helped Dr. Congcong Wang to finish the statistical analysis and the corresponding figures; Dr. Qihua Pang contribute to the method developments and parts of manuscript's revision. Dr. Ruifang Fan is the PI of our lab and she is in charge of the whole study and the revision of manuscript.

Conflict of interest statement

On behalf of the authors of this paper, I hereby certify that all actual or potential competing financial interests have been declared and that the authors' freedom to design, conduct, interpret, and publish research is not compromised by any controlling sponsor as a condition of review and publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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